سیستم هویت دیجیتال ملی ایران

Iran National Digital Identity System (INDIS)

سند نیازمندی‌های محصول — Product Requirements Document

نسخه / Version: 1.1

IranProsperityProject.org

طبقه‌بندی / Classification: Strategic Planning — Public Draft

زبان / Language: فارسی (Persian) + English (Bilingual)

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⚠️ **یادداشت بلاکچین / Blockchain Notice**
The blockchain layer has **not yet been selected**. This document treats the distributed ledger as a **pluggable component**. All architecture is designed to support **Hyperledger Fabric** as the primary candidate, with requirements written to avoid vendor lock-in. The final selection should follow a formal technical evaluation. See Section 5.3 for details.

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PART I: Executive Summary and Vision

بخش اول: خلاصه اجرایی و چشم‌انداز

1.1 Document Purpose / هدف سند

This Product Requirements Document defines the complete technical, functional, and non-functional requirements for the **Iran National Digital Identity System (INDIS)** — a sovereign digital identity infrastructure designed to serve as the foundational trust layer for post-transition Iran.

این سند نیازمندی‌های کامل فنی، کارکردی و غیرکارکردی **سیستم هویت دیجیتال ملی ایران (INDIS)** — را تعریف می‌کند زیرساختی برای هویت دیجیتال ملی‌حاکمیتی که به‌عنوان لایه اعتماد بنیادی برای ایران پس از گذار طراحی شده است.

Intended Audience / مخاطبان هدف:
- Technical architects and software engineers / معماران فناوری و مهندسان نرم‌افزار
- Government policy stakeholders / ذینفعان سیاست‌گذاری دولتی
- International development and audit partners / شرکای توسعه و ممیزی بین‌المللی
- Security auditors and privacy researchers / ممیزان امنیتی و پژوهشگران حریم خصوصی

1.2 Vision Statement / بیانیه چشم‌انداز

INDIS will be the most **privacy-respecting**, **cryptographically verifiable**, and **institutionally trustworthy** national identity system in the Middle East and Central Asia — exceeding the UAE Pass model across three dimensions:

Dimension	UAE Pass	INDIS Target
Privacy Architecture	Centralised data store	ZK-proof selective disclosure; citizen-held keys
Trust Model	Government-centralised	Blockchain-anchored, decentralised verification
Inclusivity	Urban smartphone-first	Offline-capable, multilingual, social attestation pathway
Language Support	Arabic + English	**Persian-first RTL**, 8 languages, minority dialect support
Post-Quantum Security	Not addressed	NIST PQC standards for long-term credentials

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1.3 Strategic Objectives / اهداف راهبردی

The system must achieve the following outcomes within its first **180 days** of phased deployment, aligned directly with the Emergency Phase Booklet:

Objective	Booklet Alignment	Success Metric
Referendum Readiness	Political Chapter — referendum within 4 months	Verified voter roll operational before referendum date
Pension Ghost Elimination	Macroeconomic Chapter — fiscal stabilisation	Ghost beneficiary rate reduced by ≥80% within 90 days
Personnel Vetting	Military Chapter — vetting by Day 40	100% of priority personnel credentialed by Day 40
Justice Infrastructure	Transitional Justice — Truth Commission	Anonymous testimony system operational by Day 60 (Phase 2)
Service Continuity	Government Essential Functions	Single auth layer replacing fragmented legacy systems

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1.4 Guiding Principles / اصول راهنما

Every design decision is governed by the following principles in **strict priority order**:

1. **Privacy by Architecture** — Privacy is a mathematical guarantee, not a policy promise
2. **Sovereignty First** — No foreign entity has administrative access to Iranian citizen data
3. **Inclusion Without Exception** — Every Iranian has a path to enrollment regardless of geography, literacy, or documentation status
4. **Adversarial Security** — Designed assuming active subversion attempts by regime remnants and foreign actors
5. **Transparent System, Private Citizens** — Code and protocols are public; citizen data is not
6. **فارسی اول / Persian First** — Every interface designed RTL and Persian as the primary experience

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PART II: Stakeholder Analysis

بخش دوم: تحلیل ذینفعان

2.1 Primary User Categories / دسته‌بندی کاربران اصلی

INDIS serves **five primary user categories**, each with distinct needs, technical capabilities, and trust relationships.

2.1.1 Category A: Individual Citizens / شهروندان

Estimated Population: 88 million residents + 8–10 million diaspora

Sub-Segments / زیرگروه‌ها

Urban Digitally Literate Citizens / شهروندان شهری با سواد دیجیتال

Attribute	Detail
Smartphone ownership	~70%
Primary language	Persian
Secondary languages	Varies by ethnicity
Primary concerns	Privacy, ease of use, speed
Expected usage	Daily — banking, healthcare, government services
Technical literacy	Medium to High

Rural and Semi-Urban Citizens / شهروندان روستایی و نیمه‌شهری

Attribute	Detail
Connectivity	Limited; feature phones common
Literacy	Variable; functional illiteracy present
Regional languages	Kurdish, Azerbaijani, Arabic, Baluchi, Gilaki, Mazandarani
Primary concerns	Physical access to enrollment, not being excluded
Special requirement	Offline enrollment capability is non-negotiable

Elderly Citizens / شهروندان سالمند

Attribute	Detail
Digital literacy	Low across most of segment
Dependency	High on pension and healthcare systems
Documentation status	May have outdated or damaged documents
Design requirement	Simplified mode, assisted enrollment, audio guidance

Youth and Students / جوانان و دانشجویان

Attribute	Detail
Digital literacy	Highest of all segments
Critical role	Referendum and election participation
Primary concern	Government surveillance; data privacy
Design requirement	Transparent privacy controls prominently displayed

Undocumented / Documentation-Deficient Citizens / شهروندان فاقد مدارک

Attribute	Detail
Geographic concentration	Border regions, ethnic minority communities
Cause	Systematic exclusion from civil registration under Islamic Republic
Enrollment pathway	Social attestation (mandatory, not optional)
Political importance	Their inclusion is **non-negotiable** for transition legitimacy

Iranian Diaspora / ایرانیان خارج از کشور

Attribute	Detail
Estimated size	8–10 million in 50+ countries
Critical role	Referendum participation, professional reintegration
Enrollment pathway	Embassy and consulate network
Technical literacy	Generally high
Special requirement	Remote enrollment; multilingual support

2.1.2 Category B: Government Entities / نهادهای دولتی

Ministry / Body	Primary INDIS Use Case	Credential Types Accessed
Ministry of Interior	Civil registration, elections	Citizenship, Voter Eligibility
Ministry of Justice	Court records, criminal history	Full Identity (Level 3)
Ministry of Health	Coverage verification, prescriptions	Health Insurance
Ministry of Finance	Tax identity, pensions	Pension, Citizenship
Ministry of Education	Student enrollment, teacher credentialing	Professional, Age Range
Ministry of Foreign Affairs	Passport management	Citizenship, Diaspora
Transitional Mehestan	Member credentialing, voter rolls	Voter Eligibility
Transitional Divan	Proceeding authentication, evidence custody	Full Identity, Amnesty Applicant
NISS / Army / Police	Personnel vetting, access control	Security Clearance
Central Bank	Beneficial ownership, AML	Citizenship, Full Identity (Level 3)

2.1.3 Category C: Verifiers / تأییدکنندگان

Verifiers confirm aspects of a citizen's identity **without necessarily seeing full identity data**. This is where ZK-proof technology delivers maximum value.

Verification Tiers / سطوح تأیید

Level	Method	Use Case	Data Revealed to Verifier
Level 1	QR scan + ZK proof	General service eligibility	Boolean result only
Level 2	NFC + biometric match	Financial transactions, voting	Credential category + validity status
Level 3	Full identity check	Border control, court proceedings	Full identity **with explicit citizen consent**
Level 4	Emergency override	Critical security situations	Full identity; requires senior auth + automatic audit alert

Private Sector Verifiers / تأییدکنندگان بخش خصوصی
- Banks and financial institutions (KYC)
- Employers (work authorization)
- Insurance companies
- Telecommunications providers
- Real estate registries

International Verifiers / تأییدکنندگان بین‌المللی
- Foreign embassies (visa issuance)
- International organizations operating in Iran
- Foreign financial institutions (correspondent banking)

2.1.4 Category D: Enrollment Agents / عاملان ثبت‌نام

Agent Type	Location	Connectivity	Authority Level
Fixed Center Agents	Provincial offices, post offices, hospitals	High	Full enrollment
Mobile Unit Agents	Rural areas, border regions	Satellite/ cellular	Full enrollment
Embassy Agents	50+ countries	Variable	Full enrollment (diaspora)
Community Agents	Remote villages	Minimal	Initiate only; biometric required for completion

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2.1.5 Category E: Administrators and Auditors / مدیران و ممیزان

Role	Access Level	Restrictions
NIA System Administrators	System configuration, key ceremony	Multi-party authorization for sensitive ops
Independent International Auditors	Read-only system logs	No access to citizen data
Parliamentary Oversight Committee	Aggregate statistics, audit reports	No individual citizen data
Security Red Team	Full penetration testing	Scheduled; isolated environment

---

2.2 User Journey Maps / نقشه‌های سفر کاربر

Journey 1: Urban Citizen — Self-Enrollment

```
┌──────────────────────────────────────────────────────────────┐
│  PRE-ENROLLMENT / ثبت‌نام از پیش                              │
└──────────────────────────────────────────────────────────────┘
                          │
                          │
                          ▼
            Receives SMS/broadcast notification
            رسانه یا پیامک طریق از اطلاع‌رسانی
                          │
                          ▼
            Downloads INDIS app (iOS/Android/HarmonyOS)
            برنامه دانلود INDIS
                          │
                          ▼
            Selects language — Persian default (RTL)
            فارسی پیش‌فرض — زبان انتخاب
                          │
                          ▼
            Reads privacy notice — explicit consent captured
            صریح رضایت ثبت — خصوصی حریم اطلاعیه مطالعه
                          │
┌──────────────────────────▼───────────────────────────────────┐
│  DOCUMENT CAPTURE / مدارک ثبت                                 │
└──────────────────────────────────────────────────────────────┘
                          │
                          │
                          ▼
            Photographs existing documents (any combination):
            موجود مدارک از عکس‌برداری (قبول قابل ترکیبی هر):
            • Kart-e Melli (ملی کارت)
            • Shenasnameh (شناسنامه)
            • Passport / گذرنامه
            • Driver's License / گواهینامه
                          │
                          ▼
            AI-assisted document authenticity check
            مصنوعی هوش با مدارک صحت بررسی
                          │
┌──────────────────────────▼───────────────────────────────────┐
│  BIOMETRIC CAPTURE / بیومتریک ثبت                             │
└──────────────────────────────────────────────────────────────┘
                          │
                          │
                          ▼
            Facial recognition + liveness detection
            بودن زنده تشخیص + چهره تشخیص
                          │
                          ▼
            10-finger fingerprint capture
            انگشت ۱۰ انگشت اثر
                          │
                          ▼
```

```
            Optional iris scan
            عنبیه اسکن (اختیاری)
                        |
┌───────────────────────▼───────────────────────────┐
|  PROCESSING / پردازش                              |
└───────────────────────┬───────────────────────────┘
                        |
                        |
                        ▼
            National biometric deduplication check
            ملی سطح در بیومتریک نبودن تکراری بررسی
            ⏱️ Target: <90 seconds / هدف: ۹۰ از کمتر ثانیه
                        |
                        ▼
            DID generated ON-DEVICE
            کاربر دستگاه روی DID تولید
            🔑 Private key NEVER leaves device
            🔑 نمی‌کند ترک را دستگاه هرگز خصوصی کلید
                        |
┌───────────────────────▼───────────────────────────┐
|  CREDENTIAL ISSUANCE / دیجیتال مدارک صدور        |
└───────────────────────┬───────────────────────────┘
                        |
                        |
                        ▼
            Government issues Verifiable Credentials:
            دولت توسط تأییدپذیر مدارک صدور:
            ✓ Citizenship Credential / شهروندی مدرک
            ✓ Age Range Credential / سنی محدوده مدرک
            ✓ Residency Credential / سکونت محل مدرک
            ✓ Voter Eligibility Credential / رأی شرایط واجد مدرک
                        |
                        ▼
            Credentials stored in device wallet (encrypted)
            (رمزنگاری‌شده) دستگاه پول کیف در مدارک ذخیره
                        |
                        ▼
            Digital ID card generated + physical card option
            فیزیکی کارت گزینه + دیجیتال هویت کارت تولید
                        |
                        ▼
            ✅ ENROLLMENT COMPLETE / شد کامل ثبت‌نام
```

Journey 2: Rural Undocumented Citizen — Social Attestation

```
┌──────────────────────────────────────────────────────────────┐
│  COMMUNITY AWARENESS / آگاهی‌سازی اجتماعی                    │
└──────────────────────────────────────────────────────────────┘
                          │
                          │
                          ▼
            Mobile enrollment unit arrives in village
            می‌رسد روستا به ثبت‌نام سیار واحد
                          │
                          ▼
            Agent explains process in local language
            with visual aids (no literacy assumed)
            محلی زبان در عامل توسط فرایند توضیح
            (سواد فرض بدون) تصویری کمک‌های با
                          │
┌──────────────────────────▼───────────────────────────────────┐
│  SOCIAL ATTESTATION / اجتماعی تأیید                          │
└──────────────────────────────────────────────────────────────┘
                          │
                          │
                          ▼
            3 enrolled community members co-attest identity
            می‌کنند تأیید را هویت جامعه ثبت‌نام‌شده عضو ۳
            Each attestation cryptographically signed
            می‌شود امضا رمزنگاری به‌صورت تأیید هر
                          │
                          ▼
            Basic information captured:
            می‌شود ثبت پایه اطلاعات:
            • Name as known in community / شناخته‌شده نام
            • Approximate age / تقریبی سن
            • Village/region of origin / منطقه/روستا
            • Family relationships if known / خانوادگی روابط
                          │
                          ▼
            Biometric capture (same as standard)
            بیومتریک ثبت (استاندارد روش مانند)
                          │
                          ▼
            "Social Attestation" credential issued
            می‌شود صادر "تأیید اجتماعی" مدرک
            [Lower privilege; upgrade path displayed]
            [می‌شود داده نمایش ارتقا مسیر محدود؛ دسترسی]
                          │
                          ▼
            ✅ Basic services accessible immediately
            ✅ است دسترس در بلافاصله پایه خدمات
            📋 Voting eligibility pending secondary review
            📋 ثانویه بررسی انتظار در بودن رأی شرایط واجد
```

Journey 3: ZK-Proof Verification at Service Counter

```
 CITIZEN DEVICE            SERVICE COUNTER          BLOCKCHAIN
شهروند دستگاه            خدمات پیشخوان            بلاکچین
     |                        |                        |
     |   Verifier sends       |                        |
     |◄── verification ────────|                        |
     |    request             |                        |
     |    [Credential type    |                        |
     |     + criteria only]   |                        |
     |                        |                        |
Citizen reviews              |                        |
approval screen              |                        |
صفحه شهروند                  |                        |
می‌بیند را تأیید             |                        |
     |                        |                        |
ZK circuit loads             |                        |
credential — generates       |                        |
proof ON-DEVICE              |                        |
اثبات تولید ZK               |                        |
دستگاه روی                   |                        |
⚡ No personal data           |                        |
   ever transmitted          |                        |
     |                        |                        |
     |   ZK Proof only        |                        |
     |────────────────────────►|                        |
     |   (zero personal data) |                        |
     |                        |  Proof verification    |
     |                        |────────────────────────►|
     |                        |                        |
     |                        |  ✓ Valid / ✗ Invalid   |
     |                        |◄────────────────────────|
     |                        |                        |
     |                  ✅ or ❌ displayed              |
     |                  NO citizen data                |
     |                  seen by verifier               |
     |                  شخصی داده هیچ                  |
     |                  نمی‌شود داده نمایش             |
```

PART III: Functional Requirements

بخش سوم: نیازمندی‌های کارکردی

3.1 Core Identity Management / مدیریت هویت مرکزی

FR-001: Enrollment Processing / پردازش ثبت‌نام

ID	Requirement	Priority
FR-001.1	System SHALL process enrollment via three pathways: Standard (documents + biometrics), Enhanced (civil registry + biometrics), Social Attestation (community + biometrics)	MUST
FR-001.2	Biometric deduplication SHALL complete within **90 seconds** under normal load	MUST
FR-001.3	System SHALL generate a DID conforming to **W3C DID Core 1.0** for each enrolled individual	MUST
FR-001.4	Private keys SHALL be generated **on citizen's device only**. Government servers SHALL NEVER hold citizen private keys	MUST
FR-001.5	System SHALL support minor enrollment through parent/guardian with linked guardian credential	MUST
FR-001.6	System SHALL issue a **temporary enrollment receipt** credential immediately after biometric capture	MUST
FR-001.7	System SHALL support **bulk enrollment** for military, civil service, and hospital populations	SHOULD
FR-001.8	Social attestation enrollment SHALL require minimum 3 co-attestors, each holding a valid INDIS credential of Tier 2 or above	MUST
FR-001.9	Diaspora enrollment SHALL be available through the embassy network with full credential parity	MUST

FR-002: Credential Types / انواع مدارک دیجیتال

Credential Type	نام فارسی	Issuing Authority	Validity	Key Attributes
Citizenship	مدرک شهروندی	National Identity Authority	Permanent	Status, country-of-birth range, reg. date
Age Range	مدرک محدوده سنی	NIA	Permanent	Bracket only: <18, 18–30, 31–50, 51–70, >70
Voter Eligibility	مدرک واجد شرایط رأی	Electoral Commission	Per election cycle	Eligibility boolean, district code
Residency	مدرک اقامت	Ministry of Interior	Annual	Province code, urban/rural
Professional	مدرک حرفه‌ای	Relevant Ministry	Defined by issuer	Category, qualification level
Health Insurance	مدرک بیمه سلامت	Ministry of Health	Annual	Coverage type, enrollment status
Pension	مدرک بازنشستگی	Ministry of Finance	Monthly	Beneficiary status, payment eligibility
Security Clearance	مدرک مجوز امنیتی	NIA / NISS	90-day	Clearance level only (no details)
Amnesty Applicant	مدرک متقاضی عفو	Transitional Divan	Proceeding duration	Application status, case reference
Diaspora	مدرک ایرانی خارج از کشور	MFA	Annual	Diaspora status, country of residence
Social Attestation	مدرک تأیید اجتماعی	NIA + Community	1 year (renewable)	Attestation level, upgrade path

FR-002 Rules:
- **FR-002.R1:** Revocation propagation to all verifier nodes: ≤ 60 seconds
- **FR-002.R2:** Revocation registry SHALL be on-chain, checkable without querying central identity DB
- **FR-002.R3:** Selective disclosure MUST be supported for all credential types
- **FR-002.R4:** Expiry notifications: 30 days, 7 days, 1 day before expiry
- **FR-002.R5:** Credential delegation SHALL be supported with full audit trail

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FR-003: Zero-Knowledge Proof System / سیستم اثبات دانش صفر

Architecture Decision: ZKPs are the privacy backbone of INDIS. They are not optional features.

Proof Systems by Use Case

Use Case	ZK System	Rationale
Standard credential verification	**ZK-SNARKs (Groth16)**	Fast proof generation (<3s on mid-range phone)
Electoral / referendum verification	**ZK-STARKs**	Post-quantum security; no trusted setup required
Batch credential operations	**PLONK**	Universal trusted setup; efficient for bulk ops
Anonymous testimony (Justice)	**Bulletproofs**	No trusted setup; range proofs for age/date

Standard ZK Proof Operations

Age and Identity Proofs:

```
prove_age_above(threshold: u8) → Proof
  // Proves age ≥ threshold without revealing exact age

prove_age_in_range(min: u8, max: u8) → Proof
  // Proves age within range without revealing exact age

prove_citizenship() → Proof
  // Proves Iranian citizenship without revealing any identifier

prove_voter_eligibility(election_id: Hash) → Proof
  // Atomic proof: citizenship + age ≥ 18 + not in exclusion list
  // Reveals: nothing beyond eligibility boolean
```

Credential Validity Proofs:

```
prove_credential_valid(credential_type: CredType) → Proof
  // Proves: issued by authorized issuer AND not revoked AND not expired

prove_credential_issued_within(start: Date, end: Date) → Proof
  // Proves issuance date within range without revealing exact date
```

Selective Attribute Proofs:

```
prove_attribute_meets_criteria(
  credential: Credential,
  attribute: AttributeKey,
  predicate: Predicate
) → Proof
  // Proves attribute satisfies predicate without revealing attribute value
```

FR-003 Performance Targets:

Operation	Target	Maximum
ZK proof generation (standard)	2 seconds	5 seconds
ZK proof generation (electoral STARK)	5 seconds	15 seconds
Proof verification at terminal	200ms	500ms
Trusted setup ceremony participation	Multi-party; public	—

FR-003 Governance Requirements:
- ALL ZK circuit code SHALL be **open source and publicly audited**
- Trusted setup (SNARK) SHALL be **multi-party computation** with international observers
- Formal verification of ZK circuits SHALL be completed before production deployment
- Circuit audit reports SHALL be published publicly

FR-004: Biometric Management / مدیریت بیومتریک

Modality	Required	Minimum Acceptable	Fallback
Facial recognition	Yes	1 face	Audio-guided for visually impaired
Fingerprint	Yes	4 fingers	If injury prevents full capture
Iris scan	Optional	N/A	Not required
Voice recognition	Optional	N/A	Accessibility pathway

Performance Requirements:

Metric	Target	Standard
False Match Rate	≤ 0.0001%	ISO/IEC 29794
False Non-Match Rate	≤ 0.1%	ISO/IEC 29794
Liveness Detection (IAPAR)	≤ 0.5%	ISO/IEC 30107-3
Presentation Attack Detection	Mandatory	ISO/IEC 30107-3

FR-004 Privacy Rules:
- Biometric templates stored ONLY in encrypted form (AES-256-GCM, HSM-managed keys)
- One-way transformation applied before storage — original cannot be reconstructed
- Biometric data SHALL NEVER be shared with any foreign government, organization, or private entity
- Alternative pathways mandatory for persons with disabilities or occupational biometric wear

---

3.2 Citizen Application / برنامه شهروندی

FR-005: Platform Support / پشتیبانی از سکوها

Platform	Minimum Version	Notes
Android	8.0 (API 26)	Primary; highest Iranian market share
iOS	14.0	Secondary
HarmonyOS	2.0	Required for Huawei devices common in Iran
Progressive Web App	Modern browsers	Fallback for unsupported devices
USSD / SMS	Feature phones	Essential for rural low-connectivity access

FR-006: Language and RTL Support / پشتیبانی از زبان و راست‌به‌چپ

CRITICAL: Persian is designed first. All other language interfaces are derived from the Persian/RTL design, not the reverse.

Language	Script	Direction	Priority	Status
فارسی (Persian)	**Naskh / Nastaliq**	**RTL**	**Primary**	**Launch**
English	Latin	LTR	**Co-Primary**	**Launch**
کردی سورانی (Kurdish Sorani)	Modified Arabic	RTL	Tier 1	Launch
کردی کرمانجی (Kurdish Kurmanji)	Latin	LTR	Tier 1	Launch
آذربایجانی (Azerbaijani Turkish)	Latin / Arabic	LTR / RTL	Tier 1	Launch
عربی (Arabic — Iranian dialect)	Arabic	RTL	Tier 1	Launch
بلوچی (Baluchi)	Arabic	RTL	Tier 1	Phase 2
فرانسه (French)	Latin	LTR	Tier 2	Phase 3

Localization Technical Requirements:

Requirement	Specification
Calendar system	Solar Hijri (Shamsi) default; Gregorian secondary
Number display	Persian numerals (۰۱۲۳۴۵۶۷۸۹) default; switchable
Date format	روز/ماه/سال with full Persian month names
Typography	**Vazirmatn** (open-source, comprehensive Unicode, excellent mobile legibility)
RTL cursor	Correct right-to-left cursor movement
Bidirectional text	Correct mixed Persian-English rendering (Unicode Bidi algorithm)
Search/sort	Persian alphabetical order (الفبا), not Unicode code point order
Name handling	Supports Arabic-origin names, compound names, names without family names

FR-007: Digital Identity Card Display / نمایش کارت هویت دیجیتال

The primary screen SHALL display a digital identity card styled in **Iranian cultural heritage visual language**.

Card Display Elements:

```
┌──────────────────────────────────────────────────┐
│  ایران ملی دیجیتال هویت سیستم                    │
│  Iran National Digital Identity System           │
├──────────────────────────────────────────────────┤
│                                                  │
│   [PHOTO]    نام ،خانوادگی نام                   │
│              Family Name, Given Name             │
│                                                  │
│              ملی کد: ●●●●●●●●●● [نمایش]          │
│              National ID: ●●●●●●●●●● [Reveal]    │
│                                                  │
│   ✓ شهروندی  ✓ دهنده‌رأی  ✓ سلامت بیمه          │
│   ✓ Citizen  ✓ Voter      ✓ Health Ins.          │
│                                                  │
│   [QR — tap biometric to reveal]                 │
│   [کیوآر — لمس بیومتریک برای نمایش]              │
│                                                  │
│   وضعیت: تأییدشده ✅  |  Status: Verified ✅     │
└──────────────────────────────────────────────────┘
```

FR-008: Privacy Control Center / مرکز کنترل حریم خصوصی

The Privacy Control Center SHALL be a **prominently accessible** feature — not buried in settings. This is a core political commitment.

Required Displays:
- Complete history of all verification requests received
- Complete history of all credentials shared
- List of verifier categories that accessed credentials in past 12 months
- Data minimization settings per service category
- Per-verifier sharing rules: Always Share / Always Ask (default) / Never Share

Citizen Rights Enforced by UI:
- Real-time alert for every credential verification attempt
- One-tap decline for any verification request
- Complete data export (cryptographically signed package, 72-hour delivery)
- Data correction request with tracked resolution
- Escalation pathway to an independent ombudsman

3.3 Government Portal / درگاه دولتی

FR-009: Ministry Integration Dashboard

Requirement	Detail
Authentication	Certificate-based (no password-only access)
Data minimization	Role-based data visibility; pension officer CANNOT see security clearance
Audit logging	Every data access logged; retained 10 years minimum
Bulk operations	Require senior approval for operations affecting >1,000 records
Approval chains	Multi-level, configurable per credential type

FR-010: Electoral Authority Module / ماژول هیأت انتخابات

This module has a hard deadline: operational before the constitutional referendum (≤4 months post-transition)

Property	Implementation
Completeness	Every valid vote counted — cryptographic proof
Soundness	No invalid vote counted — ZK-STARK verification
Privacy	No one can determine how any individual voted
Individual Verifiability	Every voter can verify their vote was counted
Universal Verifiability	Any mathematician can verify total result
Receipt-Freeness	Cannot prove to third party how you voted (anti-coercion)

Both **in-person** (QR scan at polling station) and **remote voting** (cryptographic ballot) SHALL be supported.

FR-011: Transitional Justice Module / ماژول عدالت انتقالی

Anonymous Testimony System:
- Witness proves Iranian citizenship (ZK proof) without revealing identity
- Cryptographic receipt issued on submission
- Follow-up testimony linkable to original without identity disclosure
- Optional identity reveal to sealed court record only, with explicit consent

Conditional Amnesty Workflow:
- Verified full identity required (applicant cannot use ZK anonymity)

- Victim notification system (notifies victims of application without revealing applicant identity prematurely)
- Multi-party review with mandatory recusal checking
- Additional encryption layer; keys in judicial multi-party escrow

---

3.4 Audit, Physical Card, USSD, and Accessibility / ممیزی، کارت فیزیکی، USSD و دسترس‌پذیری

FR-014: Audit Service / سرویس ممیزی

The audit service provides tamper-evident logging for all identity and credential operations.

ID	Requirement	Priority
FR-014.1	Every identity operation (register, resolve, deactivate) SHALL generate an immutable audit entry	MUST
FR-014.2	Audit entries SHALL be cryptographically hash-chained: each entry's hash incorporates the previous entry's hash	MUST
FR-014.3	Audit entries SHALL record: action type, actor DID, subject DID, resource ID, timestamp, IP range	MUST
FR-014.4	Audit log SHALL be append-only at the database level (no UPDATE or DELETE permitted)	MUST
FR-014.5	Parliamentary Oversight Committee SHALL have read-only access to aggregate audit reports with no citizen PII	MUST
FR-014.6	Audit entries SHALL be queryable by: time range, actor DID, subject DID, action category	SHOULD
FR-014.7	Audit log integrity SHALL be verifiable by any party holding the public hash chain	MUST

FR-015: USSD / SMS Fallback / پیامک / USSD مسیر پشتیبان

For feature phone and no-connectivity access — essential for rural coverage.

ID	Requirement	Priority
FR-015.1	System SHALL expose a USSD short code for basic credential status checks (e.g., `*ID#`)	MUST
FR-015.2	USSD flow SHALL verify voter eligibility: citizen enters national ID fragment + PIN, receives YES/NO	MUST
FR-015.3	USSD flow SHALL support pension payment eligibility check	MUST
FR-015.4	SMS OTP SHALL be supported as an authentication factor for low-capability devices	MUST
FR-015.5	USSD flows SHALL be available in Persian, Kurdish, Azerbaijani, Arabic	SHOULD
FR-015.6	USSD sessions SHALL NOT retain any citizen data after session end	MUST

FR-016: Physical Identity Card / کارت هویت فیزیکی

ID	Requirement	Priority
FR-016.1	Physical cards SHALL conform to **ICAO 9303** Machine Readable Travel Documents specification	MUST
FR-016.2	Cards SHALL include an **ISO 7816** contact chip and **ISO 14443** NFC interface	MUST
FR-016.3	Chip SHALL store the holder's DID document and public key only — no biometric raw data	MUST
FR-016.4	Offline verification of card SHALL be possible using a card-reader terminal with cached NIA public key	MUST
FR-016.5	Card SHALL display Persian name (Nastaliq), national ID (masked), photo, and validity date	MUST
FR-016.6	Card SHALL include a printed QR code for INDIS app scanning	SHOULD
FR-016.7	First issuance SHALL be free; replacement fee subject to policy decision (see Open Question 5)	MUST

FR-017: Accessibility / دسترس‌پذیری

ID	Requirement	Priority
FR-017.1	Citizen app SHALL comply with **WCAG 2.1 Level AA**	MUST
FR-017.2	Audio guidance SHALL be provided for all enrollment steps (screen-reader and dedicated audio mode)	MUST
FR-017.3	Alternative biometric pathways SHALL exist for persons with physical disabilities preventing standard capture	MUST
FR-017.4	Font sizes SHALL be adjustable; minimum body text 16sp on mobile	MUST
FR-017.5	Simplified enrollment mode SHALL exist for low-literacy users (icon-driven, minimal text)	SHOULD
FR-017.6	Assisted enrollment by an authorized agent SHALL be supported for users unable to self-enroll	MUST

3.5 Verifier Application / برنامه تأییدکننده

FR-012: Verifier Registration and Authorization

Every verifier organization SHALL register with the NIA and receive a **verifier certificate** defining:
- Which credential types it is authorized to request
- Which attributes within credentials it may verify
- Maximum verification frequency limits
- Permitted verification contexts
- Geographic scope (nationwide / provincial / site-specific)

FR-013: Verification Result Display

CRITICAL: Verifier terminals SHALL display ONLY a binary result for ZK-proof verifications.

```
APPROVED                          DENIED
شد تأیید                          شد رد

[GREEN SCREEN]                    [RED SCREEN]
✅                                ❌

Voter eligibility: CONFIRMED      Voter eligibility: NOT CONFIRMED
شد تأیید :رأی شرایط واجد         نشد تأیید :رأی شرایط واجد

No citizen data displayed.        No reason displayed to verifier.
هیچ شخصی داده نمایش شودنمی داده  شودنمی داده نمایش تأییدکننده به دلیل
```

PART IV: Non-Functional Requirements

بخش چهارم: نیازمندی‌های غیرکارکردی

4.1 Performance Requirements / نیازمندی‌های عملکردی

Operation	Target	Maximum	Notes
Credential verification (online)	500ms	2s	—
Credential verification (cached offline)	100ms	500ms	Up to 72h offline
ZK proof generation (standard Groth16)	2s	5s	On 2020 mid-range Android
ZK proof generation (electoral STARK)	5s	15s	—
Proof verification at terminal	200ms	500ms	—
Biometric deduplication	30s	90s	Full national population
Blockchain write finality	1s	3s	—
App launch to ready state	2s	5s	Cold start
Credential issuance	5s	30s	—
Bulk enrollment processing	500K/day	—	Peak campaign
Electoral event verification	2M/hour	—	Referendum day

4.2 Availability Requirements / نیازمندی‌های دسترس‌پذیری

Service Tier	Uptime SLA	Max Downtime/Year	RPO	RTO
Core identity verification	99.99%	53 minutes	15 min	1 hour
Enrollment services	99.9%	8.7 hours	1 hour	4 hours
Electoral services (active election)	99.999%	5 minutes	5 min	30 min
Non-critical reporting	99.5%	44 hours	4 hours	8 hours

4.3 Security Requirements / نیازمندی‌های امنیتی

Cryptographic Standards

Component	Standard	Notes
Data in transit	TLS 1.3 minimum	TLS 1.2 only as legacy fallback
Data at rest	AES-256-GCM	—
Digital signatures	Ed25519 or ECDSA P-256	—
Key management	FIPS 140-2 Level 3 HSM	Or equivalent
Long-term credentials	NIST PQC (CRYSTALS-Dilithium)	Post-quantum readiness
Cryptographic libraries	Audited open-source ONLY	No proprietary crypto
ZK circuits	Formally verified	Before production

Threat Model

Threat	Attack Vector	Mitigation
Regime Remnant Manipulation	Corrupt agents creating fraudulent identities for persons evading accountability	Agent credential signing; pattern anomaly detection; mandatory supervisory review for social attestation
Foreign State Cyberattack	Nation-state attack on electoral infrastructure	Air-gapped backups; distributed architecture; international security partnerships; regular red-teaming
Identity Fraud at Scale	Multiple identity enrollment for vote manipulation or benefit fraud	Biometric deduplication at enrollment; liveness detection; cross-database reconciliation
Future Government Overreach	Next authoritarian government using INDIS for mass surveillance	ZK-proof architecture makes surveillance technically impossible at protocol level; citizen-controlled private keys; parliamentary oversight with technical enforcement
Insider Threat	NIA administrators abusing privileged access	RBAC with minimum privilege; all access logged; multi-party authorization for sensitive ops; separation of duties

Audit and Testing Requirements

Requirement	Specification
Pre-launch security audits	Minimum 2 independent internationally recognized firms
Automated security scanning	Continuous; results published monthly
Bug bounty program	Public; established before production launch
Formal ZK circuit verification	Required before deployment
Red team exercises	Quarterly; scenarios include regime-remnant and nation-state
Penetration testing	Annual; scope covers all user-facing surfaces

4.4 Privacy Requirements / نیازمندی‌های حریم خصوصی

Requirement	Detail
Data Minimization	Collect only strictly necessary data per function
Purpose Limitation	Enrollment data used ONLY for identity verification — not law enforcement without judicial order
Cross-Verifier Correlation	Architecture SHALL make cross-verifier behavioral profiling technically impossible without judicial authorization
Differential Privacy	Aggregate population statistics use differential privacy (ε parameter published publicly)
Biometric Sovereignty	Biometric data NEVER shared with any foreign entity under any circumstances
DPIA	Published before each major component goes to production

4.5 Scalability and Load Testing Requirements / نیازمندی‌های مقیاس‌پذیری و تست بار

Milestone	Load Target	Test Requirement
Phase 1 readiness	2M enrolled; 100K verifications/day	Load test at 3× peak before launch
Phase 2 (referendum)	10M enrolled; 2M verifications/hour on election day	Load test at 5× peak; red team on electoral module
Phase 3 national rollout	50M enrolled; 500K bulk enrollments/day	Sustained 48-hour load test
Phase 4 full coverage	88M enrolled; 1M verifications/hour sustained	Annual capacity review

Load Testing Requirements:

- All performance targets in §4.1 SHALL be validated by automated load tests before each phase launch
- Load tests SHALL be run against a production-equivalent environment (same hardware class, same data volume)
- Load test results SHALL be published publicly before Phase 2 (electoral) launch
- Biometric deduplication SHALL be tested at full national population scale (88M) in an isolated environment before Phase 3

4.6 Offline and Low-Connectivity Requirements

Capability	Detail
Citizen app offline	Full credential presentation and ZK proof generation without network
Verifier offline	Up to 72 hours using cached revocation lists
Enrollment agent offline	Complete capture with queued sync
SMS fallback	USSD / SMS short codes for basic eligibility checks on feature phones
Physical card	ISO 7816 embedded chip; offline verification via card-reader terminals

PART V: System Architecture

بخش پنجم: معماری سیستم

5.1 High-Level Architecture

```
╔═══════════════════════════════════════════════════════════════════╗
╔═══════════════════════════════════════════════════════════════════╗
╔═══════════════════════════════════════════════════════════════════╗
```

```
│  │  Audit Log     │  │  • No personal data   │  │  Key Mgmt    │  │
│  │  Append-only   │  └───────────────────────┘  │  HSM Cluster │  │
│  │  Crypto-signed │                              │  FIPS 140-2  │  │
│  └────────────────┘                              └──────────────┘  │
└────────────────────────────────────────────────────────────────────┘
```

5.2 ZK Proof Data Flow

```
┌──────────────────────────────────────────────────────────────────────┐
│              ZK PROOF GENERATION AND VERIFICATION FLOW               │
│         صفر دانش اثبات تأیید و تولید جریان                          │
└──────────────────────────────────────────────────────────────────────┘

1. CREDENTIAL STORAGE (at enrollment / نام‌ثبت هنگام)
   ────────────────────────────────────────────────────
   Government issues VC → NIA signs with private key
                        → Credential anchored on blockchain (hash only)
                        → Credential delivered to citizen device
                        → Stored in encrypted device wallet
                        → NIA private key NEVER on citizen device
                        → Citizen private key NEVER on government server

2. VERIFICATION REQUEST (at service point / خدمت نقطه در)
   ──────────────────────────────────────────────────────
   Verifier terminal → Sends: CredentialType + Predicate + Nonce
                       e.g., "Prove voter eligibility for election #7"
                       NO citizen data sent in request

3. ZK PROOF GENERATION (on citizen device / شهروند دستگاه روی)
   ──────────────────────────────────────────────────────────────
   Device loads ZK circuit for requested proof type
   Circuit inputs: [credential, predicate, nonce]
   Circuit outputs: [proof] — contains ZERO personal data
   Generation time: <5 seconds
   Citizen approves sharing → Proof sent to verifier

4. VERIFICATION (at verifier + blockchain / تأیید)
   ────────────────────────────────────────────────────
   Verifier receives proof
   Checks proof against:
     (a) ZK verification key (public, downloaded from NIA)
     (b) Revocation status (queried from blockchain)
     (c) Credential anchor (queried from blockchain)
   Result: ✅ Valid or ❌ Invalid
   NO citizen data processed by verifier at any step

5. AUDIT (citizen-controlled / شهروند کنترل)
   ──────────────────────────────────────────────
   Verification event logged in citizen's own wallet
   Citizen sees: date, verifier category, credential used
   Verifier sees: boolean result only
   NIA sees: anonymized aggregate statistics only
```

5.3 Blockchain Abstraction Layer

> ⚠️ **The blockchain provider has not been selected.** This section defines the abstraction layer that decouples application logic from the specific blockchain implementation, enabling Hyperledger Fabric adoption or substitution with minimal application-layer changes.

Abstraction Interface

```
 // Blockchain Abstraction Interface
// All blockchain interactions go through this interface ONLY
// No application service SHALL call blockchain SDK directly

interface BlockchainAdapter {

  // DID Operations
  registerDID(did: string, document: DIDDocument): Promise<TxReceipt>
  resolveDID(did: string): Promise<DIDDocument>
  updateDIDDocument(did: string, update: Partial<DIDDocument>): Promise<TxReceipt>
  deactivateDID(did: string): Promise<TxReceipt>

  // Credential Anchoring
  anchorCredential(credentialHash: Hash, issuerDID: string): Promise<TxReceipt>
  verifyAnchor(credentialHash: Hash): Promise<AnchorStatus>

  // Revocation Registry
  revokeCredential(credentialId: string, reason: RevocationReason):
Promise<TxReceipt>
  checkRevocationStatus(credentialId: string): Promise<RevocationStatus>
  getRevocationList(issuerDID: string): Promise<RevocationList>

  // Audit Trail (anonymized)
  logVerificationEvent(event: AnonymizedVerificationEvent): Promise<TxReceipt>

  // Health and Status
  getBlockHeight(): Promise<number>
  getValidatorStatus(): Promise<ValidatorStatus[]>
  estimateTxTime(): Promise<milliseconds>
}
```

Hyperledger Fabric Implementation Notes

When Hyperledger Fabric is selected, the following design decisions apply:

Decision	Specification
Network topology	Private permissioned; NIA as ordering service admin
Consensus	Raft (CFT) for performance; PBFT if Byzantine fault tolerance required
Channel structure	Separate channels per credential type for data isolation
Chaincode language	**Go** (primary) + TypeScript (secondary)
Identity MSP	Fabric CA integrated with NIA PKI
State database	**CouchDB** for rich queries on credential status
Block time	Target: 500ms; Maximum: 2 seconds
Endorsement policy	Minimum 3-of-5 NIA nodes for credential operations
Peer node distribution	Minimum 21 nodes; geographically distributed
Privacy extensions	Fabric Private Data Collections for sensitive credential metadata
No personal data on-chain	ENFORCED by chaincode — hash only; plaintext rejected

Blockchain Selection Evaluation Criteria

The final selection should evaluate candidate platforms against:

Criterion	Weight	Notes
Iranian sovereignty (no foreign control)	25%	Critical — must eliminate vendor admin access
Throughput (TPS at production load)	20%	Minimum 10,000 TPS for election day
Finality time	15%	Target <3 seconds
HSM integration	15%	FIPS 140-2 Level 3 required
Open source maturity	10%	Active community; auditable
Operational complexity	10%	Must be operable by Iranian technical teams
International standards compliance	5%	W3C DID, VC Data Model

Current Candidate Ranking:

Platform	Sovereignty	TPS	Finality	Recommendation
Hyperledger Fabric	✅ Full control	~3,500 base; higher with tuning	<2s	**Primary candidate**
Hyperledger Besu	✅ Full control	~1,000-2,000	2-5s	Secondary candidate
Custom BFT (purpose-built)	✅ Full control	Highest	<1s	Long-term option post-stabilisation
Public Ethereum / L2	❌ No sovereignty	Variable	Variable	Not suitable

---

PART VI: Technology Stack

بخش ششم: پشته فناوری

6.1 Stack Overview

```
┌──────────────────────────────────────────────────────────────────┐
│                    INDIS TECHNOLOGY STACK                        │
│              ایران ملی هویت سیستم فناوری پشته                    │
├────────────────────┼─────────────────────────────────────────────┤
│  LAYER / لایه      │  TECHNOLOGY CHOICES                         │
├────────────────────┼─────────────────────────────────────────────┤
│ Mobile (iOS)       │ Swift / SwiftUI                             │
│ Mobile (Android)   │ Kotlin / Jetpack Compose                    │
│ Mobile (Harmony)   │ ArkTS / ArkUI                               │
│ Mobile (PWA)       │ React + TypeScript (RTL-first)              │
├────────────────────┼─────────────────────────────────────────────┤
│ ZK Circuits        │ Circom 2.0 + SnarkJS (Groth16/PLONK)        │
│                    │ StarkWare Cairo (STARKs for elections)      │
│                    │ Bulletproofs-rs (anonymous testimony)       │
├────────────────────┼─────────────────────────────────────────────┤
│ Backend Services   │ Go (primary — performance-critical services) │
│                    │ Rust (ZK proof service, crypto operations)  │
│                    │ Python (ML/AI services, data analytics)     │
├────────────────────┼─────────────────────────────────────────────┤
│ API Layer          │ gRPC (internal service communication)       │
│                    │ REST + OpenAPI 3.0 (external integrations)  │
│                    │ GraphQL (government portal queries)         │
├────────────────────┼─────────────────────────────────────────────┤
│ API Gateway        │ Kong (open-source) or NGINX + custom        │
│                    │ mTLS for all service-to-service calls       │
├────────────────────┼─────────────────────────────────────────────┤
│ Identity DB        │ PostgreSQL 16 (primary relational store)    │
│                    │ TimescaleDB extension (audit time-series)   │
│                    │ Redis (session cache, revocation cache)     │
├────────────────────┼─────────────────────────────────────────────┤
│ Biometric DB       │ PostgreSQL + pgvector (template storage)    │
│                    │ Air-gapped deployment; HSM key management   │
│                    │ Neurotechnology SDK or open NIST-compliant  │
├────────────────────┼─────────────────────────────────────────────┤
│ Blockchain         │ [TBD] — Hyperledger Fabric primary candidate│
│                    │ Abstraction layer enforced (see §5.3)       │
│                    │ CouchDB state DB (Fabric)                   │
├────────────────────┼─────────────────────────────────────────────┤
│ Key Management     │ HashiCorp Vault + HSM backend               │
│                    │ AWS CloudHSM / Thales Luna (if partnership) │
│                    │ Custom national HSM if sovereignty requires  │
├────────────────────┼─────────────────────────────────────────────┤
│ Infrastructure     │ Kubernetes (container orchestration)        │
│                    │ Helm (deployment management)                │
│                    │ Terraform (infrastructure as code)          │
│                    │ Istio service mesh (mTLS, observability)    │
├────────────────────┼─────────────────────────────────────────────┤
│ Observability      │ Prometheus + Grafana (metrics)              │
│                    │ OpenTelemetry (distributed tracing)         │
│                    │ ELK Stack (log aggregation — privacy-safe)  │
```

```
├──────────────────┼──────────────────────────────────────────┤
│ CI/CD            │ GitLab CI (self-hosted)                  │
│                  │ ArgoCD (GitOps deployment)               │
│                  │ SonarQube (code quality)                 │
│                  │ Trivy (container security scanning)      │
├──────────────────┼──────────────────────────────────────────┤
│ AI / ML          │ Python + PyTorch (biometric deduplication) │
│                  │ ONNX Runtime (on-device inference)       │
│                  │ scikit-learn (fraud pattern detection)   │
│                  │ Apache Kafka (real-time event streaming) │
├──────────────────┼──────────────────────────────────────────┤
│ Biometric SDK    │ NIST BOZORTH3 (fingerprint matching)     │
│                  │ OpenFace / ArcFace (facial recognition)  │
│                  │ ISO 30107-3 compliant liveness detection │
├──────────────────┼──────────────────────────────────────────┤
│ Standards        │ W3C DID Core 1.0                         │
│                  │ W3C Verifiable Credentials 2.0           │
│                  │ OpenID Connect 4 Verifiable Presentations │
│                  │ ISO/IEC 18013-5 (mDL — mobile ID)        │
│                  │ ICAO 9303 (physical card)                │
│                  │ ISO/IEC 30107-3 (PAD / liveness)         │
│                  │ FIPS 140-2 Level 3 (HSM)                 │
│                  │ NIST PQC (post-quantum)                  │
└──────────────────┴──────────────────────────────────────────┘
```

6.2 Technology Rationale / دلایل انتخاب فناوری

Why Go for Backend Services?

Go provides the right balance of performance, concurrency (critical for biometric deduplication under load), and operational simplicity. The standard library is comprehensive, deployment is a single binary, and the language is well-suited for building high-reliability government infrastructure. The Iranian developer community has growing Go expertise.

Why Rust for ZK and Crypto?

Cryptographic operations and ZK proof generation are memory-safety-critical. Rust's ownership model eliminates entire classes of vulnerabilities (buffer overflows, use-after-free) that are catastrophic in cryptographic code. The Rust ZK ecosystem (arkworks, bellman, bulletproofs-rs) is mature and actively audited.

Why Circom + SnarkJS for ZK Circuits?

Circom is purpose-built for ZK circuit development with strong tooling and a large audited circuit library. SnarkJS provides browser and Node.js proving, enabling both server-side and on-device proof generation from a single circuit codebase.

Why Vazirmatn for Persian Typography?

- Open-source (OFL license) — no licensing costs or restrictions
- Complete Unicode coverage including all Persian, Arabic, and extended characters
- Specifically optimized for screen rendering at small sizes on mobile devices
- Maintained by the Iranian open-source community
- Available in multiple weights suitable for both display and body text

Why Self-Hosted GitLab CI?

All source code, CI/CD pipelines, and deployment tooling must be under Iranian sovereign control. Cloud-hosted services (GitHub Actions, CircleCI) create foreign dependencies incompatible with the sovereignty requirement.

6.3 What We Are NOT Using

Technology	Reason Excluded
AWS / Azure / Google Cloud	Foreign government subpoena risk; sovereignty violation
Any closed-source cryptographic library	Cannot be audited; trust cannot be established
Public blockchains (Ethereum, Solana, etc.)	No sovereignty; unpredictable costs; foreign validator control
Biometric SDKs without source audit rights	Cannot verify absence of backdoors
Any technology with Iranian sanctions exposure	Legal risk to international technical partners

PART VII: Implementation Roadmap

بخش هفتم: نقشه راه پیاده‌سازی

7.1 Phase Overview

```
MONTH:  1    2    3    4    5    6    7    8    9   10   11   12   18   24
        |    |    |    |    |    |    |    |    |    |    |    |    |    |
        ├────┤                                                            |
PHASE 0 |────| Foundation and governance                                  |
        ├────┴────┤                                                       |
PHASE 1         |────────| Priority enrollment                           |
                ├────────┤                                                |
PHASE 2         |  Hard deadline: referendum ready ◄──────────────────────|
                |         |                                               |
                ├─────────┴──────────────────────┤                        |
PHASE 3                                          |────────────| National rollout
                                                 ├────────────┴──────────┤
PHASE 4                                                                  |───
                                                                Full coverage|
```

7.2 Phase 0 — Foundation (Months 1–2)

Governance Establishment:

- [ ] Establish National Identity Authority (NIA) as governing body
- [ ] Conduct ZK trusted setup ceremony with international observers (SNARK)
- [ ] Open-source publication of all core cryptographic code
- [ ] Publish Data Protection Impact Assessment (DPIA)
- [ ] Establish Parliamentary Oversight Committee with technical staff
- [ ] Publish public bug bounty program
- [ ] Initiate blockchain platform evaluation and selection process

Technical Foundation:

- [ ] Deploy blockchain validator nodes (minimum 21, geographically distributed)
- [ ] Establish HSM cluster for key management (FIPS 140-2 Level 3)
- [ ] Deploy core identity and credential services in staging environment
- [ ] Complete initial penetration testing of core infrastructure
- [ ] Deploy CI/CD pipeline (self-hosted GitLab)
- [ ] Establish DR sites and test failover procedures

Deliverables:

- Functioning staging environment

- Published security audit results
- Governance framework operational
- Blockchain platform selected OR evaluation criteria published with timeline

---

7.3 Phase 1 — Priority Enrollment (Months 2–3)

Target Population: Military/security personnel, civil servants, healthcare workers
Hard Requirement: Military vetting operational by Day 40 (Emergency Phase Booklet — Military Chapter)
Scale Target: 2 million enrollments

Technical Milestones:
- [ ] Mobile enrollment application: iOS + Android in Persian (RTL-first) and English
- [ ] Enrollment agent application: 50+ fixed enrollment centers
- [ ] Biometric deduplication operational for up to 2 million population
- [ ] Initial credential types live: Citizenship, Age Range, Security Clearance
- [ ] Government portal operational for Ministry of Interior and Defense
- [ ] Bulk enrollment pipeline operational

Definition of Done:
- 100% of defined priority personnel enrolled and credentialed
- Security clearance credential issued and verifiable
- Zero critical security findings in production environment

7.4 Phase 2 — Electoral Preparation + Justice Foundation (Months 3–4)

> ⚠️ **HARD DEADLINES:**
> - **Day 60:** Anonymous testimony system operational (Transitional Justice — Truth Commission, §1.3)
> - **Month 4:** Electoral module operational and independently audited for constitutional referendum

Technical Milestones:
- [ ] Electoral module: fully operational and independently audited (STARK-ZK)
- [ ] Voter eligibility credential: issued to all Phase 1 enrolled citizens
- [ ] Remote voting capability: deployed and penetration tested
- [ ] International observer access tools: deployed
- [ ] SMS/USSD verification fallback: operational for low-connectivity regions
- [ ] Diaspora voting portal: operational through embassy network
- [ ] **[Day 60]** Anonymous testimony system: operational (ZK citizenship proof + receipt tokens)
- [ ] **[Day 60]** Conditional amnesty workflow: operational (full identity required, judicial escrow)

Electoral System Audit Requirements:
- Independent audit by minimum 2 internationally recognized firms
- MUST complete minimum 14 days before referendum date
- Audit reports published publicly
- International observer participation in verification process
- End-to-end verifiability demonstrated publicly before election day

Scale Target: 10 million enrollments (full urban coverage)

7.5 Phase 3 — National Rollout (Months 4–12)

Technical Milestones:
- [ ] Mobile enrollment units: deployed to all 31 provinces
- [ ] All credential types: operational
- [ ] Pension and subsidy payment integration: complete
- [ ] Healthcare and pharmacy verification: deployed
- [ ] Full minority language support: Kurdish, Azerbaijani, Arabic, Baluchi
- [ ] Transitional Justice module: operational (Truth Commission + Amnesty)
- [ ] Private sector verifier program: launched (banks, telecoms)
- [ ] Physical card distribution: nationwide

Scale Target: 50 million enrollments

7.6 Phase 4 — Full Coverage (Months 12–24)

Technical Milestones:
- [ ] Diaspora enrollment: complete through full embassy network
- [ ] Physical card distribution: complete for non-smartphone populations
- [ ] International interoperability: framework with partner countries
- [ ] Post-quantum migration: all long-term credentials migrated to CRYSTALS-Dilithium
- [ ] System optimization: based on 12 months of production telemetry
- [ ] Full open-source publication: all components

Scale Target: 85+ million enrollments (full coverage)

PART VIII: Blockchain Framework Detail

بخش هشتم: جزئیات چارچوب بلاکچین

8.1 What the Blockchain Stores (and What It Does NOT)

```
STORES ✅                          DOES NOT STORE ❌
──────────────────────────         ──────────────────────────────
DID Document (public key,          Names
  service endpoints)               Addresses
Credential hash anchors            National ID numbers
Revocation status (boolean)        Biometric data
Anonymized verification events     Any credential content
Validator node status              Photos
System health metrics              Any personal information
                                   Any linkable identifier
```

This separation is enforced at the chaincode level. Chaincode will reject any transaction containing data that matches personal information patterns.

8.2 Hyperledger Fabric Network Topology

```
┌──────────────────────────────────────────────────────────────────────┐
│              HYPERLEDGER FABRIC NETWORK TOPOLOGY                     │
│          (IF FABRIC IS SELECTED AS BLOCKCHAIN PLATFORM)              │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  ORDERING SERVICE (Raft)                                             │
│  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐               │
│  │ Orderer 1│ │ Orderer 2│ │ Orderer 3│ │ Orderer 4│               │
│  │ Tehran   │ │ Isfahan  │ │ Mashhad  │ │ Tabriz   │               │
│  └──────────┘ └──────────┘ └──────────┘ └──────────┘               │
│                                                                      │
│  PEER NODES (minimum 21, geographically distributed)                 │
│  ┌──────────────────────────────────────────────────────┐           │
│  │ Org: NIA    │ Org: MoI    │ Org: MoH   │ Org: MoF  │           │
│  │ 6 peers     │ 5 peers     │ 5 peers    │ 5 peers   │           │
│  └──────────────────────────────────────────────────────┘           │
│                                                                      │
│  CHANNELS                                                            │
│  ┌─────────────────┐  ┌─────────────────┐  ┌──────────────┐        │
│  │ did-registry    │  │ credential-     │  │ electoral    │        │
│  │ channel         │  │ anchor channel  │  │ channel      │        │
│  │ (all orgs)      │  │ (all orgs)      │  │ (NIA + MoI)  │        │
│  └─────────────────┘  └─────────────────┘  └──────────────┘        │
│                                                                      │
│  CHAINCODES (Go)                                                     │
│  • did-registry-cc     : DID CRUD operations                         │
│  • credential-anchor-cc: Hash anchor + revocation                    │
│  • audit-log-cc        : Anonymized verification events              │
│  • electoral-cc        : ZK-STARK proof anchoring                    │
│                                                                      │
│  ENDORSEMENT POLICY                                                  │
│  Credential operations: 3-of-5 NIA peer endorsement                  │
│  DID registration: 2-of-3 NIA peer endorsement                       │
│  Revocation: 2-of-2 (NIA + issuing org) endorsement                  │
│                                                                      │
│  PRIVATE DATA COLLECTIONS                                            │
│  Sensitive credential metadata shared only between                   │
│  authorized organizations; hash on public ledger only                │
└──────────────────────────────────────────────────────────────────────┘
```

8.3 Migration Strategy

If the initial blockchain selection needs to be changed (e.g., starting with a simpler implementation and migrating to Fabric later), the abstraction layer in Section 5.3 enables this without application-layer changes.

Migration Steps:

1. Deploy new blockchain network in parallel
2. Export all DID documents and credential anchors from existing chain
3. Replay all historical transactions on new chain (cryptographically verified)
4. Switch abstraction layer adapter via configuration change
5. Verify zero data loss through cryptographic reconciliation
6. Decommission old chain after 30-day verification period

PART IX: Open Questions and Decisions Required

بخش نهم: سؤالات باز و تصمیمات مورد نیاز

The following items require **policy decisions from the Transitional Government** before final technical implementation can proceed.

#	Decision Required	Options	Impact on Timeline	Recommendation
1	**Blockchain Platform Selection**	Hyperledger Fabric / Besu / Custom	±3 months if delayed past Month 1	Start Fabric evaluation immediately; select by end of Month 1
2	**Biometric Sovereignty**	Fully onshore from Day 1 / International deduplication partner during Phase 1	±3–6 months if fully onshore required	Phased: partner for Phase 1, migrate onshore by Phase 3
3	**Diaspora Voting Eligibility**	Eligible for referendum / Eligible for Mehestan elections only / Not eligible	Determines Phase 2 scope significantly	Recommended: Eligible for all; aligns with Emergency Phase Booklet
4	**Social Attestation Threshold**	3 co-attestors / 5 co-attestors / Tiered by region	Minor development impact; major policy impact	3 co-attestors minimum; 5 recommended for border regions
5	**Physical Card Fee**	Free for all / Means-tested free / Fee for all	Budget and rollout speed	Free for first issuance; fee for replacement
6	**International Audit Partners**	Estonia e-Governance Academy / UNDP / UN OCHA / Others	Must be decided before Phase 0 ends	Estonia + UNDP recommended
7	**Minority Language Launch Scope**	All 5 at launch / Persian + Kurdish + Azerbaijani at launch	~2 months development difference	Persian + 3 major regional languages at launch
8	**Data Retention After Death**	10 years / 25 years / Permanent	Database sizing and legal framework	25 years recommended; aligns with transitional justice needs
9	**Private Sector Verifier Fees**	Fee per verification / Annual license / Free	Revenue model for system sustainability	Annual license; graduated by organization size
10	**ZK Trusted Setup**	National ceremony only / International multi-party	Legitimacy and security	International multi-party; include diaspora participants
11	**API Versioning Policy**	URL versioning (/v1/, /v2/) / Header versioning / Both	Determines breaking-change contract for all integrators (banks, ministries)	URL versioning for public REST API; gRPC uses proto package versioning

#	Decision Required	Options	Impact on Timeline	Recommendation
12	**Biometric SDK Procurement**	Open-source (BOZORTH3 + ArcFace) / Commercial (Neurotechnology, Aware) with audit rights / Build in-house	±4–6 months if custom build required	Evaluate open-source first; commercial only if ISO accuracy targets unmet

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Appendix A: Glossary / واژه‌نامه

Term	فارسی	Definition
DID	شناسه غیرمتمرکز	Decentralized Identifier — W3C standard for sovereign digital identity
Verifiable Credential (VC)	مدرک تأییدپذیر	Cryptographically signed digital credential conforming to W3C VC Data Model
ZK-SNARK	—	Zero-Knowledge Succinct Non-interactive Argument of Knowledge
ZK-STARK	—	Zero-Knowledge Scalable Transparent Argument of Knowledge (post-quantum)
HSM	ماژول امنیت سخت‌افزاری	Hardware Security Module — tamper-resistant key storage
INDIS	سیستم هویت دیجیتال ملی ایران	Iran National Digital Identity System
NIA	سازمان هویت ملی	National Identity Authority — governing body for INDIS
RTL	راست‌به‌چپ	Right-to-Left text direction
Social Attestation	تأیید اجتماعی	Enrollment pathway for undocumented citizens using community co-attestors
Selective Disclosure	افشای انتخابی	Sharing only specific credential attributes, not the full credential
Liveness Detection	تشخیص زنده بودن	Biometric security preventing use of photos/videos to spoof enrollment
PAD	تشخیص حمله تقدیم	Presentation Attack Detection — ISO/IEC 30107-3

Appendix B: Key Standards and References

Standard	Scope	URL
W3C DID Core 1.0	Decentralized Identifiers	w3.org/TR/did-core
W3C VC Data Model 2.0	Verifiable Credentials	w3.org/TR/vc-data-model-2.0
OpenID4VP	Credential presentation	openid.net/specs/openid-4-verifiable-presentations
ISO/IEC 18013-5	Mobile driving licence / digital ID	iso.org
ICAO 9303	Physical travel document	icao.int
ISO/IEC 30107-3	Biometric liveness / PAD	iso.org
ISO/IEC 29794	Biometric sample quality	iso.org
FIPS 140-2 Level 3	HSM security standard	nist.gov
NIST PQC (FIPS 203/204/205)	Post-quantum cryptography	nist.gov/pqcrypto
Circom 2.0	ZK circuit language	docs.circom.io
Hyperledger Fabric	Permissioned blockchain	hyperledger-fabric.readthedocs.io

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Appendix C: Revision History

Version	Date	Author	Changes
0.1	۲۵۸۵/۰۱	Initial Draft	First complete draft
1.0	۲۵۸۵/۰۱	Review	Blockchain abstraction layer added; Hyperledger Fabric as candidate; Persian language requirements expanded; technology stack added
1.1	۲۵۸۵/۱۲	Engineering Review	Added FR-014 (Audit), FR-015 (USSD/SMS), FR-016 (Physical Card), FR-017 (Accessibility); fixed Day 60 Justice deadline alignment with Phase 2; upgraded PostgreSQL reference to v16; added §4.5 load testing NFRs; added Open Questions 11–12

این سند نسخه ۱.۰ است و پس از بررسی ذینفعان به‌روزرسانی خواهد شد.

This document represents Version 1.0 and will be updated following stakeholder review.

نسخه فارسی این سند برای اهداف حاکمیتی سند معتبر است.
The Persian language version of this document is the authoritative version for governance purposes.

کد منبع تمام اجزای رمزنگاری باید متن‌باز و قابل ممیزی عمومی باشد.
Source code for all cryptographic components must be open-source and publicly auditable.

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